DMLScope: Uncovering Logical Bugs in Database DML via Approximate State-Relation–Driven Metamorphic Testing

Abstract

Database Management Systems (DBMSs) are critical infrastructure for modern data-driven applications, and logical bugs in DBMSs may silently produce incorrect query results or corrupted database states. While prior work has made significant progress in detecting logical bugs in read-only queries, considerably less attention has been paid to state-modifying SQL statements. Data manipulation language (DML) statements such as `INSERT`, `UPDATE`, and `DELETE` directly modify persistent database contents, making their correctness harder to validate and their consequences more severe.

In this paper, we propose *Approximate State-Relation–Driven Metamorphic Testing* (ASRMT), a novel testing paradigm for detecting logical bugs in state-modifying SQL statements. ASRMT characterizes DML semantics through state effects, an abstract and comparable representation of the state changes induced by DML execution, and validates correctness by checking approximate relations between the effects of related statements. Based on this paradigm, we develop DMLScope, which systematically generates executable DML statements, synthesizes approximate variants, executes them from the same database pre-state, and verifies whether their state effects satisfy the expected approximate relations. We evaluate DMLScope on six widely used DBMSs and uncover 47 previously unknown bugs, including 21 logical bugs. The results show that DMLScope effectively detects logical bugs in DML statements and substantially broadens the scope of DBMS testing beyond existing techniques.

1 Introduction

Database Management Systems (DBMSs) are critical infrastructure for modern data-driven applications, ranging from e-commerce and online banking to large-scale enterprise systems [9, 39]. They provide standardized interfaces such as SQL for both data retrieval and data modification [6]. Given their central role, bugs in DBMSs can cause serious consequences, including incorrect query results, data corruption, and even security violations [14, 38]. Among them, logical bugs are particularly dangerous because they do not trigger crashes, but silently produce incorrect results or database states, often remaining hidden for long periods [17, 28, 32]. Recent studies have shown that even mature and widely tested DBMSs still contain many previously unknown logical bugs, highlighting the difficulty of ensuring DBMS correctness [7, 12, 16, 28–30, 32–34].

Although prior work has made substantial progress in detecting logical bugs in read-only queries [7, 12, 16, 28–30, 32–34], much less attention has been paid to state-modifying SQL statements. In this work, we focus on data manipulation language (DML) statements, including `INSERT`, `UPDATE`, `DELETE`, and DBMS-specific extensions such as `REPLACE`. Unlike read-only queries, DML statements directly modify persistent database contents, making their correctness harder to validate and their consequences more severe [32]. Moreover, DML execution often involves complex predicates, nested subqueries, joins, and integrity-related side effects, which further increase the risk of subtle logical bugs.

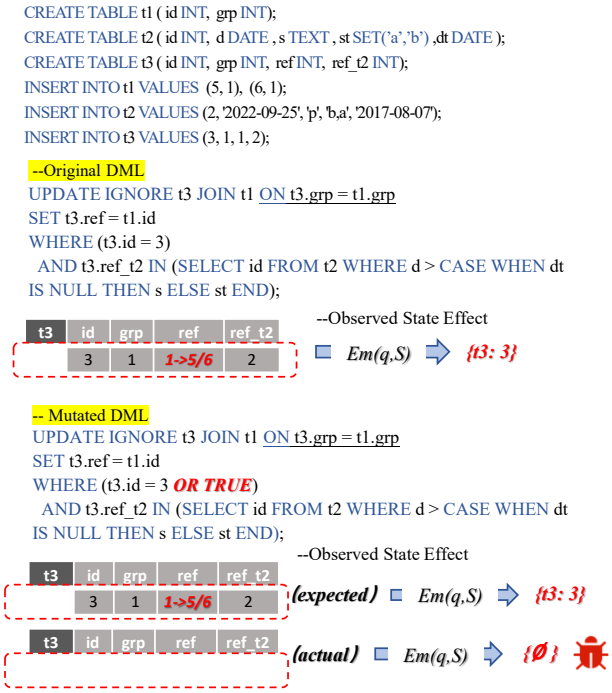

Figure 1: An example of a logical bug detected by DMLScope in MySQL (Bug #120047).

Figure 1 shows a real DML bug from MySQL (Bug #120047 [24]), where the DBMS incorrectly handles row selection during the execution of a join-based `UPDATE` statement with a complex predicate. In this example, the mutated DML joins tables `t3` and `t1`, and assigns `t1.id` to `t3.ref` for the row in `t3` that satisfies the `WHERE` clause. However, due to an internal logic error, MySQL fails to update the qualifying row and produces an incorrect final database state. This bug does not cause a crash or raise any execution error; instead, the statement appears to execute normally while silently violating its intended semantics.

Logical bugs in DBMSs, especially those in DML statements, are difficult to detect automatically. A fundamental challenge lies in the *test oracle problem* [13], i.e., determining whether the execution of a given SQL statement produces a correct result or database state. While this problem has been extensively studied for read-only queries, it becomes significantly more challenging for DML statements, as correctness must be defined over state transitions rather than query outputs. To address this challenge, recent work DQE [32] proposes a differential testing approach that detects logical bugs by comparing the rows accessed by `SELECT`, `UPDATE`, and `DELETE` statements sharing the same predicate. The key insight of DQE is that these statements should access the same set of tuples under identical conditions, and any inconsistency indicates a potential logical bug. Although effective, DQE relies on cross-statement consistency and treats DML correctness indirectly through row-access equivalence. This design introduces several limitations: (1)

it requires restricting SQL features to those commonly supported across different statement types, limiting its applicability to complex queries; (2) it cannot naturally capture value-level correctness (e.g., incorrect assignments in UPDATE); and (3) it may fail to detect bugs when multiple statements share the same erroneous execution path [12]. These limitations suggest that a more direct oracle over database state transitions is needed for DML testing.

There have also been several studies on detecting logical bugs in DBMSs [7, 12, 16, 28–30, 33, 34]. For instance, TLP [29] partitions a query based on three-valued logic and checks equivalence between decomposed queries, while NoREC [28] rewrites queries into non-optimized forms to expose inconsistencies. These techniques are effective at detecting logical bugs in read-only queries, where correctness can be defined over result sets. However, their primary focus is on query equivalence or result-set relations, rather than the correctness of state-modifying semantics. In contrast, our work focuses on logical bugs in DML statements, where errors manifest as incorrect database state transitions, including wrong tuple selection, unintended modifications, and incorrect value assignments.

Inspired by Pinolo [12], which demonstrates that approximate semantic relations can serve as effective oracles for logical bug detection, we revisit this principle in the context of state-changing SQL. Our key insight is that, although the exact post-state of a DML statement is difficult to specify as an oracle, its relative behavior can still be reliably constrained through approximation. We therefore term this paradigm ***A**pproximate **S**tate-**R**elation–Driven **M**etamorphic **T**esting* (**ASRMT**). Under ASRMT, the semantics of a DML statement are characterized by its state effect, an abstract and comparable representation of the induced state change which we formalize in Section 3. Given a DML statement and its over- or under-approximate variants, their corresponding state effects are expected to satisfy a approximate relation. Any violation of these approximate state-relation invariants indicates a potential logical bug. For example, in Figure 1, we generate a mutated UPDATE statement by transforming the predicate into a logically broader one. Under our approximation-based reasoning, the broader statement should induce a state change that subsumes that of the original statement. However, MySQL updates one row for the original DML but no row for the mutated one, thereby violating the expected inclusion relation. Such a contradiction serves as clear evidence of a logical bug.

Based on this paradigm, we propose DMLScope, a practical testing framework for detecting logical bugs in DML statements. DMLScope instantiates ASRMT through a state-oriented generate–mutate–execute–verify workflow. Starting from a valid database pre-state, it first generates executable DML statements that satisfy schema and integrity constraints. It then synthesizes over- and under-approximate variants using semantics-aware mutators with known approximation directions. To ensure meaningful comparison, DMLScope executes the original and mutated statements independently from the same pre-state under differential state execution. After execution, it extracts state effects using representations tailored to the mutation semantics, and finally checks whether the observed effects satisfy the expected approximate state relations. Any violation is reported as a potential logical bug.

We evaluate DMLScope on six widely used and extensively tested DBMSs: MySQL, MariaDB, MonetDB, Percona, DuckDB, and TiDB. Overall, DMLScope detected 21 logical bugs across all tested systems, including 5 in MySQL, 3 in MariaDB, 5 in MonetDB, 5 in Percona, 1 in DuckDB, and 2 in TiDB. In addition, it also triggered 8 abnormal errors and 18 crashes, yielding 47 unique bugs in total. Among all reported bugs, 47 have been confirmed by developers, and 27 have already been fixed. These results demonstrate the effectiveness of DMLScope in finding bugs in DBMSs. Overall, we make the following contributions:

- We propose Approximate State-Relation–Driven Metamorphic Testing, a novel testing paradigm for detecting logical bugs in state-modifying SQL statements. It characterizes DML semantics through state effects and validates correctness by checking approximate state relations over database state transitions.
- We implement DMLScope, a practical testing framework that systematically generates executable DML statements, synthesizes approximate variants, performs differential state execution, and verifies state-effect relations to detect logical bugs.
- We evaluate DMLScope on six real-world DBMSs. In total, we found 47 bugs including 21 logical bugs. To further facilitate research on DBMS testing, we open-source the tool at https://anonymous.4open.science/r/DMLScope-9F03/.

2 Background and Motivation

2.1 Background

Relational Database Management Systems. Relational Database Management Systems (DBMSs) are used to create, manage, and query relational databases [36]. In a relational DBMS, data is organized into structured tables (relations) with predefined columns and rows, enabling efficient storage, retrieval, and manipulation. Structured Query Language (SQL) is the standard language for interacting with relational DBMSs, supporting data definition, manipulation, and querying [35].

Structured Query Language. Relational DBMSs provide SQL as a unified interface for interacting with stored data, enabling a wide range of operations such as schema definition, data modification, and data retrieval. Based on their functionality, SQL statements are commonly categorized into three main types.

Data Definition Language (DDL) statements are responsible for defining and modifying database schema objects. Typical examples include CREATE TABLE, ALTER TABLE, and DROP TABLE, which are used to create new relations, adjust existing structures, and remove schema objects, respectively.

Data Manipulation Language (DML) statements are designed for modifying the data stored in database tables. Representative operations include INSERT for adding new tuples, UPDATE for modifying existing records, and DELETE for removing tuples from a table.

Data Query Language (DQL) statements are used for retrieving data from the database. The most representative DQL statement is SELECT, which supports a rich set of query constructs, including filtering conditions (e.g., WHERE), ordering (ORDER BY), aggregation functions, join operations, and grouping (GROUP BY), enabling flexible and expressive data analysis over one or multiple tables.

Although these statement types are all expressed in SQL, they differ significantly in terms of supported operators and semantics. In particular, SELECT statements provide a much richer set of relational operators and compositional capabilities, while UPDATE and DELETE

focus on state modification and typically support only a subset of these features.

DML Bugs. DML statements directly modify persistent database states, and bugs in their execution may lead to incorrect state transitions, such as updating unintended tuples, assigning incorrect values, or deleting valid data. Compared with read-only query bugs, such bugs are particularly severe because they may cause irreversible data corruption or loss. From a testing perspective, DML bugs can be broadly divided into crash bugs and logical bugs. Crash bugs manifest as explicit failures (e.g., crashes or runtime errors) and can be detected using failure-based oracles. In contrast, logical bugs do not trigger visible failures; instead, the DBMS completes execution successfully while producing an incorrect database state. These bugs are especially challenging to detect, as they require reasoning about the correctness of state changes rather than returned results. In this work, we focus on logical bugs in DML statements, as they are both practically harmful and insufficiently explored by existing DBMS testing techniques.

2.2 Limitations of Existing Works

Existing research on detecting logical bugs mainly relies on automated testing techniques. However, designing an effective testing framework is challenging due to the *test oracle problem* [13]—it is difficult to determine the correct behavior of a complex SQL statement for comparison. To address this issue, researchers have explored two main categories of approaches [16, 20, 32–34].

Differential Testing. Classical differential testing executes the same query across multiple DBMSs and compares their outputs to detect inconsistencies [5, 31]. However, dialect differences, heterogeneous semantics, and system-specific behaviors often make cross-DBMS comparison difficult in practice [12]. Different from this cross-system setting, DQE [32] brings the differential testing idea into a single DBMS by comparing multiple related statements within the same system to detect DML bugs. Specifically, it constructs `SELECT`, `UPDATE`, and `DELETE` statements sharing the same predicate, and checks whether they access the same set of tuples. However, it has several limitations. First, because its oracle requires row-access comparability across statement types, test generation must stay within the subset of SQL features that can be meaningfully aligned across `SELECT`, `UPDATE`, and `DELETE`, which limits coverage for complex queries. Second, row-access consistency only validates *which tuples* are touched, but not *how* the database state is changed; thus it cannot naturally capture value-level correctness, such as whether an `UPDATE` computes and propagates the correct new values. Third, since DQE detects disagreement rather than correctness itself, it may miss bugs when multiple related statements follow the same faulty execution path and therefore remain consistently wrong [12].

Metamorphic Testing. Another dominant line of work is metamorphic testing (MT), which constructs mutated queries from an original query and checks whether their outputs satisfy predefined semantic relations [7, 16, 28–30, 33, 34]. Representative approaches such as NoREC [28] and TLP [29] define metamorphic relations based on strict equivalence of query results. Although effective in bypassing the need for ground truth, such equivalence-based strategies constrain the search space of test cases and primarily validate surface-level correctness [12]. To overcome this limitation, Pinolo [12] extends MT by introducing approximate relations, generating over- and under-approximate queries and checking whether their results satisfy subset/superset relationships. This relaxation significantly broadens the detectable bug space and inspires more flexible oracle design. Our work is inspired by this key insight. However, existing MT techniques, including Pinolo, remain fundamentally query-result-centric: their correctness criteria are defined over returned result sets rather than over the state transitions induced by DML execution. Therefore, directly bringing approximate relations into DML testing is non-trivial. It requires not only generating semantically valid and directionally sound approximate DML statements, but also designing a new oracle that can compare their state effects rather than conventional query outputs. We elaborate on these challenges next.

2.3 Challenges

Challenge #1: Generating Complex and Valid DML Statements. To effectively expose deep logical bugs, testing must explore complex DML statements that exercise rich execution paths, such as joins, nested subqueries, and data-dependent value transformations [12]. However, prior work such as DQE [32] mainly focuses on relatively simple DML patterns, limiting its ability to cover complex semantics and deeper execution behaviors. Compared with read-only queries, generating valid and meaningful DML statements is substantially more challenging. A DML statement must not only be syntactically correct, but also satisfy various semantic constraints, including key consistency, type compatibility, source–target alignment, and integrity constraints imposed by the schema and the current database pre-state. Otherwise, the statement may fail at execution time or produce undefined state effects.

Challenge #2: Lack of State-Aware Test Oracles for DML. Unlike read-only queries, whose correctness can be defined over explicit result sets, DML statements do not produce a single observable output. Instead, their semantics are reflected in state transitions over the underlying database. This makes traditional output-based oracles inapplicable. The main difficulty is that a DML-induced state transition is both implicit and multidimensional: it determines which rows are selected as targets, which rows are actually modified, and how their values are changed. These aspects do not necessarily coincide, and some may even become unstable under realistic execution semantics. For example, in Figure 1, the target row `t3.id = 3` joins with two rows in `t1` because the join condition `t3.grp = t1.grp` is non-unique. Even if the original and mutated `UPDATE` statements both operate on the same target row, the final value written to `t3.ref` may still depend on which matching source row is chosen during execution, e.g., 5 or 6. As a result, the row-selection boundary may remain stable while the exact post-state varies, making direct comparison over full final states fragile and unreliable. Therefore, the challenge is not merely to compare whether two statements produce the same output, but to design a state-aware oracle that captures the correctness-relevant aspects of state changes while remaining robust to such execution-level ambiguity.

2.4 Solutions

Solution to Challenge #1: Structured-Guided and Constraint-Aware DML Generation. To address the difficulty of generating complex and valid DML statements, we adopt a structure-first,

constraint-guided generation strategy. We explicitly separate structural complexity from semantic validity. Specifically, we first construct structurally rich DML skeletons using a grammar that captures diverse SQL constructs, such as joins and nested subqueries, ensuring sufficient exploration of complex execution paths. We then progressively instantiate these skeletons under schema constraints and the current database pre-state, so that validity requirements—such as key consistency, type compatibility, and source–target alignment—are satisfied by construction. This design enables the systematic generation of DML statements that are both sufficiently complex and reliably executable for subsequent testing.

Solution to Challenge #2: State-Relation-Driven Metamorphic Testing. To overcome the lack of state-aware test oracles for DML, we propose ***A**pproximate **S**tate-**R**elation–Driven **M**etamorphic **T**esting* (ASRMT). Inspired by Pinolo [12], ASRMT extends approximate MT from query outputs to database state transitions. The key idea is to avoid requiring a canonical exact post-state as the test oracle. Instead, ASRMT abstracts DML execution into state effects, i.e., observable and comparable representations of how a statement changes the database state. These state effects are designed to capture the correctness-relevant aspects of state transitions, while abstracting away irrelevant execution-level differences, thereby providing a more stable basis for comparison. Based on this abstraction, we define approximate state relations as over- and under-approximation constraints over DML effects, and use them as metamorphic oracles. Finally, we execute original and mutated statements from the same pre-state and check whether their observed effects satisfy the expected approximation relations. In this way, ASRMT turns the DML oracle problem from exact post-state validation into relation checking over comparable state effects. The full model and its concrete instantiations are presented in Section 3.

2.5 Motivating Example

Figure 1 presents a real logical bug triggered by a structurally rich `UPDATE` statement that contains a join, a nested subquery, and a conditional expression. Such bugs are difficult to expose without generating complex original DML statements, since simple patterns are unlikely to exercise the relevant execution logic. Starting from the original statement, we apply ASRMT to construct a mutated variant by weakening the predicate with `OR TRUE`, thereby obtaining a logically broader statement. To detect the bug, we do not compare full final states directly, because they may be unstable under one-to-many joins; instead, we compare stable state effects that capture the correctness-relevant outcome, namely which rows are affected. Under this reasoning, the broader statement is expected to affect at least the same row as the original one, but MySQL updates no row at all, thereby revealing a logical bug.

3 Approximate State-Relation–Driven Metamorphic Testing

We now present ***A**pproximate **S**tate-**R**elation–Driven **M**etamorphic **T**esting* (ASRMT), our core paradigm for detecting logical bugs in state-modifying SQL statements. Unlike prior approximation-based techniques such as Pinolo [12], which reason about relations over query results, ASRMT defines correctness over database state transitions. The key idea is to characterize DML semantics through state effects and to validate correctness by checking approximate relations between the effects of related statements. In the following, we formalize state effects, define approximate state relations, and present the corresponding test oracle.

3.1 State Effect

Let S denote a database state, and let q be a DML statement. The execution of q over S produces a new state:

$$[\![q]\!](S) \rightarrow S'.$$

To characterize the observable behavior of a DML statement, we introduce the notion of state effect.

Definition 3.1 (State Effect)**.** Given a DML statement q and a database state S, the state effect of q over S, denoted by $E(q, S)$, is an abstract set-valued representation of the observable effect induced by executing q on S.

The notion of state effect is intentionally abstract. In our paper, it is instantiated in two forms, namely the matched-row effect and the changed-row effect, depending on which aspect of the DML semantics is to be observed.

Definition 3.2 (Matched-Row Effect)**.** Given a DML statement q and a database state S, the matched-row effect of q, denoted by $E_m(q, S)$, is the set of target-row tokens corresponding to the primary keys of rows structurally matched by q on S.

Example 3.1. Consider two tables in database state S:

$t_1(c_1, c_2) =$

c_1	c_2
1	5
2	8
3	9

$t_2(c_1, c_3) =$

c_1	c_3
1	5
2	20
3	15

Consider the DML statement:

q : **UPDATE** t_1 **JOIN** t_2 **ON** $t_1.c_1 = t_2.c_1$ **SET** $t_1.c_2 = t_2.c_3$
WHERE $t_2.c_3 > 10$

Let $PK(t_1) = \{c_1\}$ be the primary key of t_1. The set of target rows in t_1 structurally matched by q is:

$$E_m(q, S) = \{t_1{:}2,\ t_1{:}3\}.$$

Intuitively, the join and predicate select the rows in t_1 whose corresponding $t_2.c_3 > 10$, thus determining the intended target set independent of actual value updates.

The matched-row effect captures the *intended target set* of a DML statement. It is suitable when the mutation primarily changes row-selection semantics, such as predicate strengthening/weakening, join-condition mutation, or other transformations that mainly affect which target rows should be selected.

Definition 3.3 (Changed-Row Effect)**.** Given a DML statement q and a database state S, the changed-row effect of q, denoted by $E_c(q, S)$, is the set of target-row tokens corresponding to the rows whose observable states differ before and after executing q on S.

Example 3.2. Consider two tables in database state S:

$t1(c_1, c_2) =$

c_1	c_2
1	a
2	b

$t2(c_1, c_2, c_3) =$

c_1	c_2	c_3
5	x	1
6	y	2
7	z	-1

Consider the DML statement:

q : **INSERT INTO** $t1(c_1, c_2)$ **SELECT** c_1, c_2 **FROM** $t2$ **WHERE** $c_3 > 0$

The newly inserted rows correspond to primary keys $\{5, 6\}$. Let h_5 and h_6 denote their normalized row summaries. We use a normalization function $row(\cdot)$ that maps a tuple to a canonical representation (e.g., a hash of its attribute values). Then the changed-row effect is:

$$E_c(q, S) = \{t1{:}5{:}ROW(h_5),\ t1{:}6{:}ROW(h_6)\}.$$

Intuitively, the effect cannot be derived purely from structure, and must be obtained by comparing the database states before and after execution.

The changed-row effect captures the actual post-execution effect of a DML statement. It is necessary when the mutation may influence value assignments, tuple multiplicity, insertion/deletion behavior, or other effects that cannot be determined reliably from the statement structure alone.

Effect Abstraction Selection. Accordingly, the abstract state effect $E(q, S)$ used by our approach is instantiated as either $E_m(q, S)$ or $E_c(q, S)$, depending on the comparison strategy. The distinction is essential: $E_m(q, S)$ characterizes which target rows are supposed to be affected according to the statement structure, whereas $E_c(q, S)$ characterizes which rows are actually changed in the database after execution.

The choice between $E_m(q, S)$ and $E_c(q, S)$ is not determined merely by the DML type, but by the semantics targeted by the mutation and the robustness of the corresponding effect abstraction. In general, $E_c(q, S)$ is more expressive, because it reflects post-execution outcomes and can capture not only row selection but also value updates, insertions, deletions, and multiplicity-related effects. However, this stronger expressiveness may also introduce irrelevant execution-level variability. For example, in join-based `UPDATE` statements, one-to-many matches or unstable source-to-target mappings may lead to ambiguous value assignments, even when the intended target-row boundary is clear. In such cases, changed-row effects may become noisy or unreliable with respect to the mutation semantics under test. By contrast, $E_m(q, S)$ is less expressive but often more stable: it abstracts away concrete value changes and focuses only on the structurally intended target set. We next provide a concrete example to illustrate this trade-off and Section 4.6 then presents the detailed strategy for selecting the effect abstraction and for extracting state effects in practice.

Example 3.3. Consider the following tables in database state S:

$t_1(c_1, c_2) =$

c_1	c_2
1	0
2	0

$t_2(c_1, c_3) =$

c_1	c_3
1	100
1	200
2	300

Consider the statement

q : **UPDATE** t_1 **JOIN** t_2 **ON** $t_1.c_1 = t_2.c_1$ **SET** $t_1.c_2 = t_2.c_3$

Structurally, the target rows in t_1 matched by q are unambiguous:

$$E_m(q, S) = \{t_1{:}1,\ t_1{:}2\}.$$

However, the changed-row effect depends on how the one-to-many join for key $c_1 = 1$ is resolved during execution. If the value 100 is chosen, one possible changed-row effect is

$$E_c(q, S) = \{t_1{:}1{:}ROW(h_{100}),\ t_1{:}2{:}ROW(h_{300})\},$$

whereas if the value 200 is chosen, another possible changed-row effect is

$$E_c(q, S) = \{t_1{:}1{:}ROW(h_{200}),\ t_1{:}2{:}ROW(h_{300})\}.$$

This example shows that changed-row effects are more expressive but less stable, since they may incorporate assignment-level ambiguity that is irrelevant when the mutation targets only row-selection semantics. Such ambiguity may arise from DBMS-specific source-row selection policies or execution-dependent factors, such as different join evaluation orders or execution plans, which can lead the same target row to be assigned from different matching source rows. In contrast, matched-row effects remain stable because they only record the structurally intended target rows.

3.2 Approximate State Relation

To generalize approximation-based metamorphic testing to state-modifying statements, we extend the notion of approximation from query result sets to database state effects.

Definition 3.4 (Approximate State Relation)**.** Given a database state S, let q_1 and q_2 be two DML statements whose state effects are $E(q_1, S)$ and $E(q_2, S)$, respectively. We say that q_1 is the *state-level under-approximation* of q_2 over S, denoted by

$$q_1 \preceq_S q_2,$$

if and only if

$$E(q_1, S) \subseteq E(q_2, S).$$

Conversely, q_1 is the *state-level over-approximation* of q_2 over S, denoted by

$$q_1 \succeq_S q_2,$$

if and only if

$$E(q_1, S) \supseteq E(q_2, S).$$

Here, $E(q, S)$ denotes the state effect induced by executing statement q on state S, and $\subseteq$ and $\supseteq$ denote inclusion and containment relations between two effect sets.

Example 3.4. Consider a table $t(c_1, c_2)$ in database state S:

$t =$

c_1	c_2
1	10
2	20
3	30

Let the following two DML statements be defined:

q_1 : **UPDATE** t **SET** $c_2 = c_2 + 1$ **WHERE** $c_1 > 1$

q_2 : **UPDATE** t **SET** $c_2 = c_2 + 1$ **WHERE** $c_1 \geq 1$

If the state effect is instantiated as the matched-row effect, then

$$E(q_1, S) = \{t{:}2,\ t{:}3\}, \qquad E(q_2, S) = \{t{:}1,\ t{:}2,\ t{:}3\}.$$

Since

$$E(q_1, S) \subseteq E(q_2, S),$$

we have

$$q_1 \preceq_S q_2.$$

Intuitively, q_2 weakens the selection predicate and therefore affects a superset of the rows affected by q_1.

This definition lifts set-semantic approximation from query outputs to state transitions, enabling reasoning about correctness in terms of persistent database effects. To systematically construct DML statements with predictable approximation relations, we introduce approximate mutators.

Definition 3.5 (Approximate Mutator)**.** Let q be a DML statement and let n be a syntactic construct in q. An *approximate mutator* μ transforms q into a new statement $q' = \mu(q)$ by modifying n such that:

$$q \preceq_S q' \quad \text{or} \quad q \succeq_S q'.$$

The direction of the induced relation determines whether μ is an under- or over-approximate mutator.

Example 3.5. Consider the statement

$$q : \textbf{DELETE FROM } t \textbf{ WHERE } c_1 > 1.$$

Let n be the comparison predicate $c_1 > 1$, and let μ be an approximate mutator that weakens the comparison operator $>$ to $\geq$. Applying μ to q yields

$$q' = \mu(q) : \textbf{DELETE FROM } t \textbf{ WHERE } c_1 \geq 1.$$

Since this transformation enlarges the set of affected rows, we have

$$q \preceq_S q'.$$

Therefore, μ is an over-approximate mutator in this example.

Approximate mutators provide a way to generate related statements whose semantic relationships are known *a priori*, which is essential for constructing effective metamorphic tests. However, a mutation typically affects only a local component of a statement. We therefore introduce approximation propagation to characterize how local changes determine global state-effect relations.

Definition 3.6 (Approximation Propagation)**.** Let q be a DML statement and let q' be obtained by applying an approximate mutator to a local construct $n \in q$. Approximation propagation describes how the local semantic deviation induced at n determines a global approximate state relation between q and q':

$$q \preceq_S q' \quad \text{or} \quad q \succeq_S q'.$$

Example 3.6. Consider a table $t(c_1, c_2)$ in database state S:

$t =$

c_1	c_2
1	5
2	10
3	15

Consider the DML statement

$$q : \textbf{DELETE FROM } t \textbf{ WHERE } \neg(c_2 > 10).$$

If the state effect is instantiated as the matched-row effect, then the affected rows are

$$E(q, S) = \{t{:}1,\ t{:}2\}.$$

since the predicate $\neg(c_2 > 10)$ is equivalent to $c_2 \leq 10$.

Now consider a local mutation that replaces the comparison operator $>$ with $\geq$ inside the negation:

$$q' : \textbf{DELETE FROM } t \textbf{ WHERE } \neg(c_2 \geq 10).$$

Although the local mutation $> \rightarrow \geq$ weakens the comparison expression itself, the surrounding negation reverses the approximation direction. Indeed, $\neg(c_2 \geq 10)$ is equivalent to $c_2 < 10$, and thus

$$E(q', S) = \{t{:}1\}.$$

Therefore,

$$E(q', S) \subseteq E(q, S),$$

which means that q' is a state-level under-approximation of q. This example illustrates that a local mutation may induce the opposite approximation direction at the statement level when it is propagated through a negation context.

Approximation propagation connects local syntactic transformations with global semantic relations, ensuring that approximation reasoning remains sound at the level of full DML statements.

3.3 Test Oracle

Based on the above definitions, ASRMT uses inclusion relations over state effects as the test oracle.

Definition 3.7 (State-Relation Oracle)**.** Given a seed statement q and its approximate variant q', a logical bug is detected if the observed state effects violate the expected inclusion relation:

- Over-approximation: $E(q, S) \not\subseteq E(q', S)$
- Under-approximation: $E(q', S) \not\subseteq E(q, S)$

This formulation enables approximation-based reasoning over persistent database behaviors, extending metamorphic testing from query correctness to state-transition correctness.

Discussion. ASRMT differs fundamentally from prior approximate metamorphic testing framework Pinolo [12], which target read-only queries and define approximation relations over result sets. For DML statements, correctness is determined not by query outputs, but by how execution changes the underlying database state. This shift requires redefining the semantic object of approximation as state effect and reasoning about approximation in terms of state transitions. Moreover, because DML semantics involve both intended target-row selection and actual post-execution changes, ASRMT must support multiple effect abstractions and propagate local mutations under state-based semantics.

4 Approach

Based on the ASRMT model, we design DMLScope, a practical testing framework for detecting logical bugs in DML statements. DMLScope operationalizes ASRMT into an executable testing workflow. In the following, we first present the overall workflow of DMLScope and then describe each core component in detail.

4.1 Overview

Figure 2 shows the overall workflow of DMLScope. The entire process follows a state-oriented generate–mutate–execute–verify paradigm. Starting from a round-specific database pre-state, DMLScope first generates an original seed DML statement, then synthesizes its approximate variants according to the predefined mutators. Next, all statements are executed under differential state execution, where each statement is executed independently from the same pre-state and the database state is restored after each execution, so that their

resulting state effects are directly comparable. Finally, DMLScope extracts state-effect tokens and checks whether the observed effects satisfy the expected approximate state relation. Any violation of the expected relation indicates a potential logical bug.

4.2 Database Initialization

Before generating and testing DML statements, DMLScope first constructs an isolated round environment by initializing a database with a valid schema and initial data. This step is not a primary contribution of our work, but is necessary to ensure the correctness and executability of subsequent testing stages. Specifically, we first generate a relational schema consisting of multiple tables with primary key and foreign key constraints, and map logical column types to DBMS-specific physical types to ensure compatibility across different systems. We then populate each table with a small number of rows while satisfying integrity constraints such as key uniqueness and referential dependencies. The insertion order follows dependency relationships so that referenced tables are initialized before dependent ones. In addition, we maintain auxiliary metadata, such as the set of existing primary keys, to guide the generation of valid DML statements and avoid referencing non-existent tuples. This design ensures that all generated statements are executable on the constructed database state and provides a stable pre-state for subsequent differential state execution.

4.3 Original DML Generation

To systematically explore diverse state-modifying behaviors, we first generate valid and executable DML statements as testing seeds. This process consists of two stages: *Grammar-guided structure generation*, which constructs abstract syntax trees (ASTs) according to a predefined grammar, and *Schema-based instantiation*, which binds schema elements and concrete values to produce executable statements. Compared with read-only query generation, original DML generation must additionally guarantee the validity of state-modifying semantics. If a generated DML statement violates integrity constraints, such as `PRIMARY KEY`, `UNIQUE`, `NOT NULL`, or `FOREIGN KEY`, it may abort entirely or cause rollback, making its state effect undefined and therefore unusable for subsequent approximation checking. Algorithm 1 summarizes the overall procedure.

Grammar-Guided Structure Generation. Initially, the algorithm performs *grammar-guided structure generation* (Lines 1–4), which determines the abstract syntactic shape of a seed DML statement. It first selects a statement kind k (Line 2), then chooses a compatible target table T (Line 3), and finally expands the corresponding grammar rules to build an abstract statement skeleton A (Line 4). We use the compact DML grammar in Figure 3 to guide this process. Beyond basic DML forms, the grammar also supports joins, nested subqueries, common table expressions, and set operations, enabling the generation of structurally rich statements. The generator recursively samples productions to construct an AST, while bounding recursion depth and restricting highly combinatorial constructs to keep generation tractable. At this stage, only the structural skeleton is determined; concrete schema bindings and values are instantiated later.

Schema-based Instantiation. Next, the algorithm performs *schema-based instantiation*, which concretizes the abstract skeleton into an executable DML statement under the current schema and database pre-state. It begins by generating candidate target columns from the skeleton and the target-table schema (Line 6), and then filters out invalid columns that are not writable, not visible in the current scope, or incompatible with the chosen statement kind (Line 7). Based on the remaining column candidates, the algorithm generates source candidates that may provide values, bindings, or filtering contexts for the statement (Line 8), and removes unstable or incompatible sources by enforcing source-side semantic constraints (Line 9). It then materializes a consistent column–source plan (π_c, π_s) (Line 10), which maps target columns to source expressions and serves as the blueprint for the subsequent instantiation process. After the plan is fixed, the algorithm binds concrete schema elements to the skeleton (Line 11), and then incrementally instantiates key positions, non-key values, and predicates (Lines 12–14). This order is important. Key positions are instantiated first because they determine tuple identity and may immediately conflict with the current pre-state. Once key values are fixed, the algorithm instantiates non-key payloads under source, type, and nullability constraints, and finally fills predicate slots with scope-correct and semantically meaningful filtering conditions. In this way, the generator does not first produce an arbitrary statement and then attempt to repair it afterwards; instead, it progressively constructs the statement while preserving semantic consistency at each stage. The algorithm then performs DML-specific semantic checks according to the statement kind (Lines 15–22). For `INSERT` and `REPLACE`, it checks key consistency and source–payload alignment (Lines 15–17), ensuring that inserted tuples neither violate uniqueness constraints nor introduce incompatible source-to-target mappings. For `UPDATE`, it checks assignment validity (Lines 18–19), so that writable target columns, source bindings, and right-hand-side expressions remain meaningful. For `DELETE`, it checks that the predicate is non-degenerate (Lines 20–21), avoiding statements that collapse into trivial or uncontrolled deletions. These statement-specific checks are essential because, unlike read-only query generation, DML generation must additionally preserve state-modifying semantics. Finally, the algorithm performs structural checking, serialization, and execution validation (Lines 23–28). It first verifies that the instantiated AST remains structurally complete and internally consistent (Line 23), and then serializes it into a concrete SQL statement q (Line 24). The statement is accepted only if it can be executed successfully on the current pre-state S (Lines 25–26); otherwise, the generation process restarts (Lines 27–28).

Figure 4 presents a simplified example of original DML generation for an `UPDATE` statement. In this example, the generator produces an abstract skeleton with unresolved table, column, and predicate slots. During the schema-based instantiation stage, the generator first selects t_1 as the target table and t_2 as the source table, because the two tables can be naturally connected through the foreign-key relationship on column c_1. It then fills the assignment slots with $t_1.c_2$ and $t_2.c_2$, since the former is writable under `UPDATE` semantics and the latter provides a type-compatible source value. Next, it instantiates the predicate slot as $t_1.c_1 = t_2.c_1$, so that the abstract join becomes a semantically meaningful row-to-row correspondence between the two tables. As a result, the final original DML statement is structurally complete, type-consistent, and executable on the current database state.

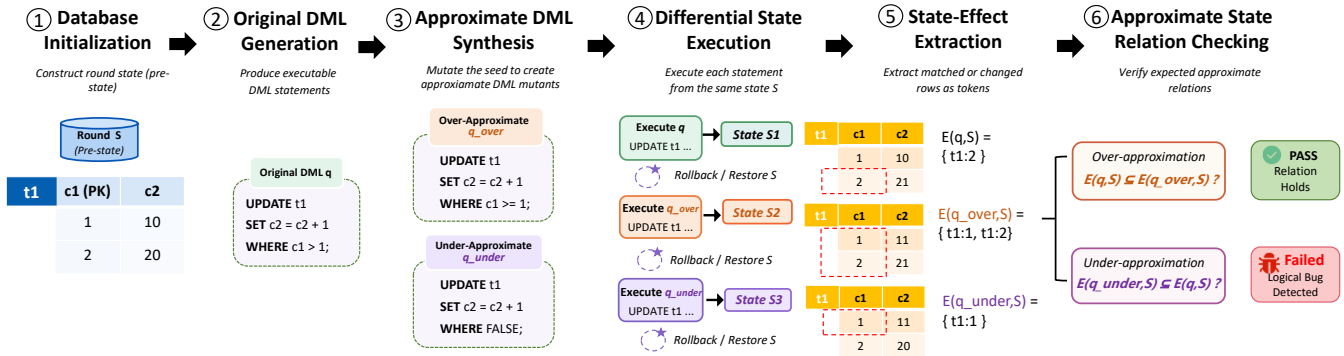

Figure 2: Overview of DMLScope.

Algorithm 1 Original DML Generation

Require: Schema $\mathcal{S}$, current database pre-state S
Ensure: Original DML statement q

// **Step 1: Grammar-guided structure generation**
Choose statement kind $k \in \mathcal{K}$, where $\mathcal{K} = \{\texttt{UPDATE}, \texttt{DELETE}, \texttt{INSERT}, \texttt{REPLACE}\}$.
Select target table $T \leftarrow \textsc{SelectTarget}(\mathcal{S}, k)$.
Build abstract skeleton $A \leftarrow \textsc{ExpandGrammar}(k, T)$.
// **Step 2: Schema-based instantiation**
Generate target-column candidates $C \leftarrow \textsc{GenColCands}(A, \mathcal{S}[T])$.
Filter invalid columns $C \leftarrow \textsc{FilterCols}(C, k, \mathcal{S}[T])$.
Generate source candidates $\mathcal{R} \leftarrow \textsc{GenSrcCands}(A, \mathcal{S}, S, C)$.
Filter unstable sources $\mathcal{R} \leftarrow \textsc{FilterSrcs}(\mathcal{R}, C)$.
Materialize a consistent plan $(\pi_c, \pi_s) \leftarrow \textsc{BindPlan}(C, \mathcal{R})$.
Bind schema elements $A \leftarrow \textsc{BindSchema}(A, \pi_c, \pi_s)$.
Instantiate key positions $A \leftarrow \textsc{InstKey}(A, S, \pi_c, \pi_s)$.
Instantiate non-key values $A \leftarrow \textsc{InstVal}(A, S, \pi_c, \pi_s)$.
Instantiate predicates $A \leftarrow \textsc{InstPred}(A, S, \pi_s)$.
if $k \in \{\texttt{INSERT}, \texttt{REPLACE}\}$ **then**
  $A \leftarrow \textsc{CheckKeyConsistency}(A, S)$
  $A \leftarrow \textsc{CheckSrcPayloadAlignment}(A, \pi_s)$
else if $k = \texttt{UPDATE}$ **then**
  $A \leftarrow \textsc{CheckAssignmentValidity}(A, \mathcal{S}[T])$
else if $k = \texttt{DELETE}$ **then**
  $A \leftarrow \textsc{CheckNonDegeneratePred}(A, \pi_s)$
end if
$A \leftarrow \textsc{CheckStructure}(A, k)$
$q \leftarrow \textsc{Serialize}(A)$.
if $\textsc{ExecValid}(q, S)$ **then**
  return q
else
  restart generation
end if

DML	D	::=	$U \mid Del \mid Ins \mid Rep$
WithClause	W	::=	WITH id AS (SQ) (, id AS (SQ))*
JoinType	J	::=	JOIN | INNER JOIN | LEFT JOIN | RIGHT JOIN
SetOp	SO	::=	UNION | UNION ALL | EXCEPT | INTERSECT
Update	U	::=	[W] UPDATE id [PARTITION(id (, id)*)] [J SRC [AS id] ON P] SET $ASSIGNS$ [FROM SRC [AS id]] WHERE P
Delete	Del	::=	[W] DELETE FROM id [PARTITION(id (, id)*)] WHERE P
		|	[W] DELETE FROM id J SRC ON P WHERE P
		|	[W] DELETE FROM id USING SRC WHERE P
Insert	Ins	::=	INSERT INTO id ($COLS$) IB
		|	INSERT INTO id ($COLS$) IB CF
Replace	Rep	::=	(REPLACE | INSERT OR REPLACE) INTO id ($COLS$) IB
InsertBody	IB	::=	VALUES ($EXPRS$) | IQ
InsertQuery	IQ	::=	$SQ \mid SS$
SelectQuery	SQ	::=	[W] SELECT [DISTINCT] $EXPRS$ FROM SRC [WHERE P] [GROUP BY $EXPRS$ [HAVING P]] [ORDER BY $EXPRS$] [LIMIT int]
SelectSet	SS	::=	$SQ \mid SQ\ SO\ SQ$
Source	SRC	::=	id [AS id]
		|	id [AS id] J (id | (SQ) | (SS)) [AS id] ON P
		|	(SQ) AS id | (SS) AS id
Predicate	P	::=	$E\ \theta\ E$ | NOT P | P AND P | P OR P | EXISTS(SQ)
		|	E IS [NOT] NULL | E [NOT] IN(SQ | $EXPRS$)
Expr	E	::=	$id \mid const \mid f(E) \mid (SS)$
Assignments	$ASSIGNS$	::=	$id = E$ (, $id = E$)*
Columns	$COLS$	::=	id (, id)*
ExprList	$EXPRS$	::=	E (, E)*
Conflict	CF	::=	ON DUPLICATE KEY UPDATE $ASSIGNS$
		|	ON CONFLICT (id) DO NOTHING
		|	ON CONFLICT (id) DO UPDATE SET $ASSIGNS$
CompOp	θ	::=	= | <> | < | > | <= | >= | LIKE | BETWEEN

Figure 3: DML Syntax Supported by DMLScope. For brevity, low-level lexical elements, including identifiers, function symbols, and constants, are abstracted as *id*, *f*, and *const*.

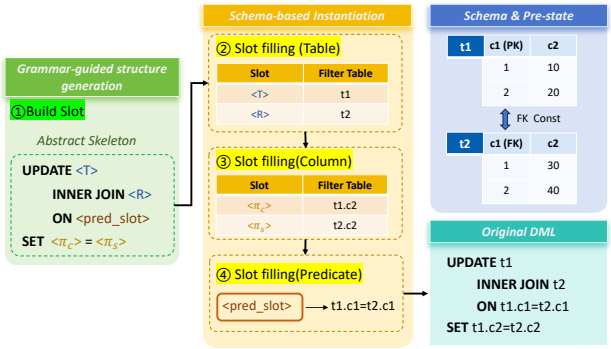

Figure 4: Illustrative Example of Original DML Generation.

4.4 Approximate DML Synthesis

Given an executable original DML statement, DMLScope next synthesizes a set of approximate variants whose state effects are expected to follow a known approximate relation with respect to the seed statement. Instead of generating new statements from scratch, the algorithm systematically applies approximate mutators to selected components of the original DML.

Table 1 summarizes some representative approximate mutators used in our synthesis phase. Our design follows the general spirit of Pinolo [12], which organizes mutators around semantic approximation, while extending it to better support DML-specific constructs. In particular, beyond relation-, predicate-, and comparison-oriented mutations, we additionally introduce *Join / Structure* mutators to handle DML-relevant constructs such as join type and HAVING-related query structure. Overall, compared with Pinolo, our framework introduces 10 additional mutators; more complete mutation rules are available in our GitHub repository [1].

Algorithm 2 summarizes the overall approximate DML synthesis procedure. Given an executable original DML statement q, the

Algorithm 2 Approximate DML Synthesis

Require: Original DML statement q
Ensure: A set of approximate variants $\mathcal{R}$
1: // **Step 1: Parse and collect mutation candidates**
2: Parse q into AST A
3: $\Gamma \leftarrow$ CollectMutableNodes(A)
4: // **Step 2: Apply approximate mutators**
5: $\mathcal{R} \leftarrow \emptyset$
6: **for all** node $n \in \Gamma$ **do**
7: $\quad\Omega \leftarrow$ ApplicableMutators(n)
8: $\quad$**for all** mutator $\mu \in \Omega$ **do**
9: $\quad\quad A' \leftarrow$ ApplyMutation(A, n, μ)
10: $\quad\quad q' \leftarrow$ Serialize(A')
11: $\quad\quad m_\ell \leftarrow$ InferLocalDirection(μ)
12: $\quad\quad m \leftarrow$ PropagateDirection(m_ℓ, n, A')
13: $\quad\quad \mathcal{R} \leftarrow \mathcal{R} \cup \{(q', m)\}$
14: $\quad$**end for**
15: **end for**
16: **return** $\mathcal{R}$

Table 1: Some representative mutators for approximate DML synthesis. Transforming the construct C_1 into C_2 achieves the under-approximation, while transforming C_2 into C_1 achieves the over-approximation.

Type	C_1	C_2
Relation	`SELECT ...` A `UNION ALL` B A `UNION` B	`SELECT DISTINCT ...` A `UNION` B A `INTERSECT` B

Predicate	P
	$P \wedge Q$
	`JOIN ... ON P`
		`FALSE`
		$P \vee Q$
		`JOIN ... ON TRUE`

Comparison	$a > b$ $a < b$ x `BETWEEN` l `AND` r	$a \geq b$ $a \leq b$ $x \geq l$

Join / Structure	`INNER JOIN`
	`HAVING P`
	`GROUP BY ... HAVING P`
		`LEFT JOIN`
		`HAVING TRUE`
		`GROUP BY ...`

algorithm first parses it into an AST and identifies a set of local constructs that admit approximate mutation (Lines 2–3), including predicates, comparison expressions, join conditions, and query-structure constructs. It then iterates over these constructs and enumerates applicable approximate mutators for each of them (Lines 6–8). For every selected mutator, the algorithm applies the mutation at the AST level and serializes the resulting structure into a concrete DML statement q' (Lines 9–10). Next, it determines the approximation direction in two stages: it first infers the local direction induced at the mutation site by the selected mutator (Line 11), and then propagates this direction through the surrounding syntactic context to obtain the global approximation direction of the mutated statement (Line 12). In particular, when the mutation site appears in direction-reversing contexts, such as `NOT`, `NOT EXISTS`, or the right branch of `EXCEPT`, the approximation direction is reversed, following the same propagation principle as in Pinolo [12].

4.5 Differential State Execution

To make state effects directly comparable across related statements, DMLScope executes the original DML statement and all its approximate variants from the same database pre-state S. This requirement is essential for ASRMT, since the expected approximate relation is defined between $E(q, S)$ and $E(q', S)$, i.e., the state effects of related statements over the same initial state.

Algorithm 3 Differential State Execution

Require: Database pre-state S, original DML statement q, approximate variants $\mathcal{R}$
Ensure: A set of post-states $\mathcal{X} = \{(q_i, S'_i)\}$
1: $\mathcal{Q} \leftarrow \{q\} \cup \{q' \mid (q', m) \in \mathcal{R}\}$
2: $\mathcal{X} \leftarrow \emptyset$
3: **for all** statement $q_i \in \mathcal{Q}$ **do**
4: $\quad$RestoreState(S)
5: $\quad S'_i \leftarrow [\![q_i]\!](S)$
6: $\quad \mathcal{X} \leftarrow \mathcal{X} \cup \{(q_i, S'_i)\}$
7: **end for**
8: **return** $\mathcal{X}$

Algorithm 3 summarizes the overall differential state execution procedure. Given a database pre-state S, an original DML statement q, and its synthesized approximate variants $\mathcal{R}$, the algorithm first materializes a statement set $\mathcal{Q}$ consisting of the original statement and all mutated statements (Line 1). It then iterates over each statement $q_i \in \mathcal{Q}$ and executes it independently from the same pre-state (Lines 3–6). Concretely, before each execution, the algorithm restores the database to S (Line 4), so that every statement observes an identical input state. It then executes the current statement to obtain its post-state S'_i (Line 5) and records this post-state as the independent execution result of q_i (Line 6). This design isolates each execution from the side effects of all others, ensuring that the post-states later used for state-effect extraction are produced from a common starting point rather than from states modified by previous executions. In practice, we implement the restoration step through dual isolation. First, each statement is executed in a separate subprocess, which isolates executions at the runtime level and prevents side effects from propagating through shared process state. Second, within each subprocess, the statement is executed inside an isolated transaction and the transaction is rolled back after execution, thereby restoring the database to the round-level pre-state S. Together, these two mechanisms ensure that every original or approximate DML statement is evaluated from the same clean execution environment. The resulting post-states are then passed to the next stage for state-effect extraction.

4.6 State-Effect Extraction

After differential state execution, DMLScope extracts state-effect tokens from each executed statement, which serve as the canonical representation for subsequent relation checking. Algorithm 4 summarizes the overall state-effect extraction procedure.

Given the execution results $\mathcal{X} = \{(q_i, S'_i)\}$ produced in Section 4.5, the algorithm processes each statement independently. For each q_i, it first parses the statement into a structured representation (Line 2), which includes its statement kind, target table, primary key, and (if available) a derived matching query (`match_sql`). For statements that admit structural matching, DMLScope derives `match_sql` from the original DML by rewriting it into a `SELECT` query that preserves its join conditions and predicates while projecting the primary keys of target rows. This query captures the row-selection semantics of the statement independent of value assignments. Based on the mutation operator and its syntactic context, the algorithm then determines an extraction strategy (Lines 3–4), which specifies both the extraction basis (`matched-row` or `changed-row`)

Algorithm 4 State-Effect Extraction

Require: Execution results $\mathcal{X} = \{(q_i, S'_i)\}$, pre-state S, mutation context (μ, n)
Ensure: State-effect token sets $\{\mathcal{T}_i\}$

1: **for all** $(q_i, S'_i) \in \mathcal{X}$ **do**
2: $(dml, match_sql) \leftarrow$ ParseDml(q_i)
3: $basis \leftarrow$ SelectBasis(μ, n, dml, S)
4: $pk_only \leftarrow$ NeedPKOnly$(\mu, n, basis)$
5: **if** $basis =$ `matched-row` **then**
6: $rows \leftarrow$ Execute$(dml.match_sql)$
7: $\mathcal{T}_i \leftarrow$ MatchToTokens$(rows, pk_only)$
8: **else**
9: $before \leftarrow$ Snapshot(S)
10: $after \leftarrow$ Snapshot(S'_i)
11: $\mathcal{T}_i \leftarrow$ DiffToTokens$(before, after, pk_only)$
12: **end if**
13: **end for**
14: **return** $\{\mathcal{T}_i\}$

and whether only primary keys should be considered. Specifically, DMLScope adopts a stability-aware selection strategy for effect extraction. It first executes the derived `match_sql` query and analyzes the mapping cardinality between source tuples and target primary keys. If multiple matched rows correspond to the same target primary key, the statement exhibits a one-to-many mapping from source-side matches to target rows. In such cases, the final value written to a target row may depend on DBMS-specific execution details, such as join order or plan selection, making the concrete state change unstable or non-deterministic. To avoid introducing such execution-level variability into the oracle, the algorithm selects the `matched-row` strategy, which focuses only on the structurally matched target-row set. Otherwise, when each target row corresponds to at most one matched source tuple, the induced value assignment is deterministic, and the algorithm selects the `changed-row` strategy to capture precise value-level state changes. The extraction follows two alternative paths depending on the selected strategy. If the `matched-row` strategy is applicable, the algorithm derives the target-row set directly from the statement structure (Lines 5–7). Specifically, it executes the derived `match_sql` query to obtain the primary keys of all structurally matched rows, and converts them into tokens of the form `table:pk`. Otherwise, the algorithm adopts the `changed-row` strategy (Lines 8–11), which extracts effects from actual state differences. It first reconstructs the pre-state snapshot S and compares it with the post-state S'_i obtained from differential execution. Rows whose observable representations differ between the two states are identified as changed rows. Each such row is normalized into a token of the form `table:pk:ROW(hash)`, capturing both row identity and value changes. Finally, all extracted tokens are normalized into a set representation, referred to as the state-effect token set, which serves as a concrete representation of the state effect $E(q_i, S)$. These token sets abstract away DBMS-specific execution details while preserving the essential semantics required for approximation checking, and are used in the next stage to verify expected over- or under-approximation relations.

Figure 5 illustrates two representative scenarios for state-effect extraction. Consider the `UPDATE` statement shown in Figure 5(a), where the mutation only affects the join predicate (e.g., $t1.c1 \leq t2.c1$). Although different executions may produce different assigned values due to multiple matching rows in $t2$, the set of affected target rows in $t1$ remains invariant. In this case, extracting

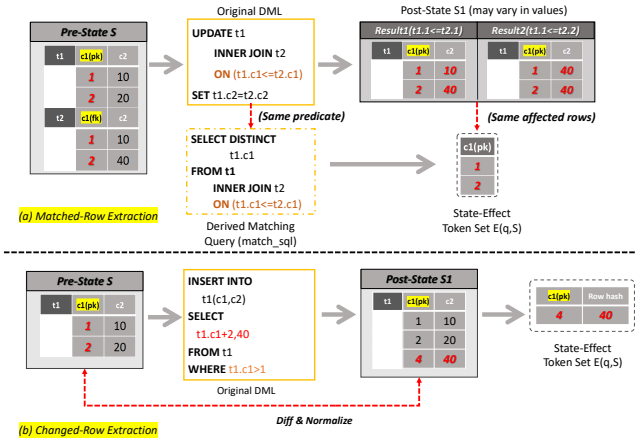

Figure 5: Illustrative Example of State-Effect Extraction.

state effects from execution results would introduce unnecessary instability. Therefore, the algorithm selects the `matched-row` strategy and derives a matching query (`match_sql`) from the statement structure. By executing this query, it obtains the set of structurally matched primary keys $\{1, 2\}$, which directly forms the state effect $E(q, S)$. In contrast, Figure 5(b) shows an `INSERT` statement, where the mutation influences both tuple generation and value assignments. Here, the affected rows cannot be determined purely from the query structure, as the inserted tuples depend on the execution outcome. Thus, the algorithm selects the `changed-row` strategy. It compares the pre-state S and the post-state $S1$ to identify newly inserted rows (e.g., $(4, 40)$), and normalizes them into tokens of the form `table:pk:ROW(hash)`. These tokens capture both row identity and value changes, yielding the state effect $E(q, S)$.

4.7 Approximate State Relation Checking

After state-effect extraction, DMLScope verifies whether the extracted token sets satisfy the expected approximate state relation. For an over-approximate mutant, the token set of the original statement should be contained in that of the mutant, whereas for an under-approximate mutant, the token set of the mutant should be contained in that of the original statement. Any violation of the expected relation indicates a potential logical bug.

4.8 Additional Implementation Details

We implemented DMLScope as a prototype DBMS testing system in Python, consisting of 13,455 lines of code. To support mutation at the structural level, DMLScope leverages SQLGlot [21] to parse original DML statements into ASTs, enabling precise application of approximate mutators while preserving syntactic and semantic correctness. During testing, DMLScope executes generated statements on target DBMSs and collects execution results for further analysis. To improve the usability of detected bugs, we adopt SQLess [19] to simplify failing cases and isolate their root causes, and further apply a deduplication strategy following Pinolo [12] to eliminate redundant bug reports. DMLScope is primarily designed for MySQL-compatible systems, including TiDB and MariaDB, but its modular architecture allows for straightforward adaptation to other DBMS dialects by customizing syntax- and type-related components. In

practice, such adaptation requires only lightweight modifications, and can be further automated using tools such as QTRAN [20], which facilitates cross-dialect deployment of MT techniques.

5 Evaluation

5.1 Experimental Setup

Tested DBMS. We evaluate DMLScope on six widely used DBMSs, as summarized in Table 2. These DBMSs were chosen based on their popularity and extensive use in real-world applications. All of these DBMSs have undergone extensive testing for both logical [7, 12, 16, 17, 20, 28–30, 32, 34, 37] and crash bugs [2, 14, 26, 38]. Although finding new bugs in such widely tested systems is challenging, it serves to validate the effectiveness of DMLScope. In our experiments, DMLScope tests the latest versions of each DBMS. We run DMLScope on each DBMS for 24 hours.

Table 2: The demographics of the DBMSs under test

DBMS	Version	GitHub Stars	DB-Engine	First Release
MySQL [23]	9.5.0	11.9K	2	1995
MariaDB [10]	12.2.2	6.7K	13	2009
MonetDB [22]	11.55.3	458	150	2004
Percona [25]	8.0.43-34	1.2K	122	2008
DuckDB [8]	1.5.0	37.3K	43	2018
TiDB [27]	8.5.5	39.4K	85	2016

Environment. We conduct the experiments on one server with 104-cores Intel(R) Xeon(R) Gold 6230R CPU @2.10GHz and 500 GB memory. The server operates under the Ubuntu 20.04 OS, with the 5.4.0-135-generic version of the Linux kernel.

5.2 Bug Detection

Table 3 summarizes the unique bugs found by DMLScope across six DBMSs. Overall, DMLScope detected 21 logical bugs, 8 abnormal errors, and 18 crashes, yielding 47 unique bugs in total. The logical bugs are distributed across all tested systems, including 5 in MySQL, 3 in MariaDB, 5 in MonetDB, 5 in Percona, 1 in DuckDB, and 2 in TiDB. Among all reported bugs, 47 have been confirmed by developers, and 27 have already been fixed. These results show that DMLScope can effectively uncover severe correctness issues even in mature and extensively tested DBMSs.

Bug Importance. Although anecdotal, developer feedback is an important indicator of an approach's effectiveness and the significance of the bugs found. Two DBMS companies reached out to us regarding our testing efforts; both were interested in how we discovered the bugs, and one even invited us to help them find more. For example, a developer from PolarDB commented: "Outstanding work, this tool looks very useful, and we will fully support any technical requirements you may need." Additionally, we received positive feedback on public bug tracking platforms. Among the confirmed bugs, all reported bugs in MariaDB were labeled as "Major," representing the highest severity level, while two bugs in MySQL and two bugs in Percona were classified as "Serious," indicating their significant impact.

Throughput. We evaluate the efficiency of our test generation by measuring the number of test cases produced per second, where each test case consists of an original DML statement and its corresponding approximate variant. Specifically, DMLScope generates

Table 3: DMLScope founded 45 unique bugs in six DBMSs.

DBMS	Bug type			Bug status	
	Logical bug	Abnormal error	Crash	Confirmed	Fixed
MySQL	5	0	3	8	0
MariaDB	3	0	0	3	0
MonetDB	5	8	12	25	25
Percona	5	0	1	6	0
DuckDB	1	0	2	3	2
TiDB	2	0	0	2	0
Total	21	8	18	47	27

xxx test cases/sec for MySQL, xxx for MariaDB, xxx for MonetDB, xxx for Percona, xxx for DuckDB, and xxx for TiDB. Beyond competitive throughput, DMLScope also attains high execution success rates on most DBMSs, reaching xxx% for MySQL, xxx% for MariaDB, xxx% for MonetDB, xxx% for Percona, xxx% for DuckDB, and xxx% for TiDB. We further measure the average complexity of generated statements using their size in bytes. The average statement size is xxx bytes for MySQL, xxx for MariaDB, xxx for MonetDB, xxx for Percona, xxx for DuckDB, and xxx for TiDB. These results show that our structure-guided and constraint-aware generation strategy can efficiently produce a large volume of complex yet executable DML statements.

Bug Diversity. We analyze the 21 confirmed logical bugs from two perspectives: the diversity of DML types and the structural complexity of the triggering statements. In terms of DML types, the majority of bugs are caused by `UPDATE` statements (15/21), while the remaining involve `DELETE` (4/21) and `INSERT` (2/21), indicating that value-modifying operations are particularly error-prone. From a structural perspective, most bugs are triggered by complex query constructs: 18 out of 21 bugs involve subqueries, 9 involve joins, 9 involve set operations, and 16 involve logic operators. These results show that DML logical bugs are highly correlated with rich query contexts and interactions among multiple SQL operators, rather than simple statements. Such complexity poses a significant challenge to existing approaches like DQE, which are limited to relatively simple and aligned query patterns and cannot effectively support complex constructs such as nested subqueries and set operations. In contrast, DMLScope can systematically generate structurally rich DML statements and leverage state-relation-based oracles to robustly reason about their effects, enabling the detection of these complex and previously overlooked logical bugs.

Case Study. Due to space limitations, we present additional representative and interesting bug cases at https://anonymous.4open.science/r/DMLScope-9F03/. ✎ [lin: add a url and false positive]

5.3 Comparative Study

To illustrate the effectiveness of our approach, we conduct our comparative study from two perspectives: tool level and oracle level. At the tool level, we compare DMLScope with six representative logical bug detection approaches, including PQS [30], NoREC [28], TLP [29], DQE [32], Pinolo [12], and EET [16], by running each tool on the latest version of each supported DBMS for 24 hours. Since the baseline methods do not support all DBMSs tested by DMLScope, the comparison is restricted to the DBMSs commonly supported by DMLScope and each baseline method. As shown in Table 5, DMLScope consistently outperforms the baselines in

Table 4: Feature distribution of detected DML logical bugs.

ID	Bug ID	Type	Subquery	Join	SetOp	LogicOp
1	MySQL#120047	UPDATE	●	●	❍	●
2	MySQL#120056	UPDATE	●	●	❍	●
3	MySQL#120057	UPDATE	●	●	●	●
4	MySQL#120058	UPDATE	❍	●	❍	●
5	MySQL#120203	UPDATE	●	❍	❍	❍
6	MariaDB#39065	DELETE	●	❍	●	❍
7	MariaDB#39097	UPDATE	●	❍	❍	●
8	MariaDB#39144	UPDATE	●	❍	●	●
9	MonetDB#7891	UPDATE	●	❍	❍	●
10	MonetDB#7888	UPDATE	❍	❍	❍	●
11	MonetDB#7889	DELETE	●	❍	●	❍
12	MonetDB#7890	INSERT	●	❍	●	●
13	MonetDB#7897	DELETE	●	❍	●	●
14	Percona#10877	UPDATE	●	●	❍	●
15	Percona#10878	UPDATE	●	●	❍	●
16	Percona#10879	UPDATE	●	●	●	●
17	Percona#10906	UPDATE	❍	●	❍	●
18	Percona#11008	UPDATE	●	❍	❍	❍
19	DuckDB#21609	INSERT	●	●	❍	●
20	TiDB#67018	UPDATE	●	❍	●	❍
21	TiDB#67019	DELETE	●	❍	●	●

Table 5: Number of logical bugs triggered in 24 hours.

DBMS	PQS	NoREC	TLP	DQE	Pinolo	EET	DMLScope
MySQL	0	-	4	0	2	3	5
MariaDB	-	0	-	0	1	-	3
MonetDB	-	-	-	-	-	-	5
Percona	-	-	-	-	-	-	5
DuckDB	-	-	-	-	-	-	1
TiDB	-	-	2	0	0	0	2
Total	0	0	6	0	3	3	21
Increment	5	3	1	10	7	4	

Note. "-" indicates that the tool does not support the DBMS.
"Increment" is computed on the commonly supported DBMSs between the baseline tool and DMLScope.

terms of triggered logical bugs, finding 5, 3, 1, 10, 7, and 4 more bugs than PQS, NoREC, TLP, DQE, Pinolo, and EET, respectively, where each increment is computed only on the commonly supported DBMSs. At the oracle level, we manually analyze whether existing testing oracles are capable of detecting the logical bugs uncovered by DMLScope. The results show that the 21 bugs detected by DMLScope are entirely disjoint from the 12 logical bugs triggered by the baseline tools. From a theoretical perspective, the 21 logical bugs detected by DMLScope fundamentally rely on reasoning about state transitions, including which rows are affected and how their values are modified. However, existing query-centric approaches, such as PQS, NoREC, TLP, Pinolo, and EET, define correctness over returned query results and therefore cannot capture errors that only manifest in persistent state changes. Although DQE targets DML statements, its oracle is based on cross-statement row-access consistency and mainly checks whether related statements access the same tuples, making it insufficient for detecting value-level errors and many complex state transitions induced by joins, subqueries, and other rich DML constructs. As a result, the logical bugs detected by DMLScope are not observable under the oracles used by these baseline approaches. This highlights that DMLScope explores a different and previously untested space of DML correctness.

6 Related Work

Logical Bug Detection in DBMSs. A variety of approaches have been proposed to detect logical bugs in DBMSs [4, 7, 12, 15, 16, 28–30, 32–34]. For example, NoREC [28] transforms an optimizable query into a non-optimizable form and detects semantic inconsistencies by comparing their outputs. TLP [29] divides query predicates into multiple subqueries and verifies that the union of their results is equivalent to the original query. EET [16] applies expression-preserving transformations, and Radar [33] compares query results between databases with and without metadata to identify semantic flaws. As discussed in Section 1 and Section 2, these approaches mainly focus on validating query correctness through returned results, such as equivalence, containment, or cross-query consistency. In contrast, DMLScope focuses on the effects of DML execution on database states. Rather than reasoning about returned result sets, it examines whether state-modifying operations produce the expected effects. From this perspective, DMLScope is largely orthogonal to existing query-centric testing techniques, as it addresses a complementary aspect of DBMS correctness.

DBMS Test-Case Generation. A variety of approaches have been proposed for generating diverse test cases for DBMSs, with the aim of improving coverage and revealing potential bugs. These techniques [2, 11, 14, 20, 38] typically focus on generating valid and varied SQL queries, but do not specifically target logical bugs. For example, SQLsmith [2] is a grammar-based DBMS fuzzer that embeds SQL grammar rules to generate complex SQL queries. SQUIRREL [38] is a mutation-based DBMS fuzzer which introduces an intermediate representation for SQL queries and models dependencies between SQL statements, enabling the generation of queries that contain multiple SQL operations. QTRAN [20] is a LLM-based approach that can automatically translate test cases from other DBMSs. These techniques aim to prevent semantic errors and improve the diversity of generated queries. In addition to general query generation, some approaches have been specifically designed to aid in bug detection. SQLRight [18] leverages code coverage feedback to enhance test-case generation. QPG [3] takes a different approach by recording the query plans covered during DBMS testing and prioritizing the mutation of queries that trigger new query plans. Different from these works, DMLScope does not primarily target general-purpose SQL generation or coverage improvement. Instead, it focuses on generating complex and semantically valid DML seeds and approximate DML variants for testing state-modifying semantics. Its main goal is not merely to increase query diversity, but to expose subtle logical bugs in DML execution by checking whether state effects satisfy expected semantic relations.

7 Conclusion

In this paper, we propose Approximate State-Relation–Driven Metamorphic Testing, a novel testing paradigm for detecting logical bugs in state-modifying SQL statements. It characterizes DML semantics through state effects and validates correctness by checking approximate relations between the effects of related statements. Based on this paradigm, we develop DMLScope, which systematically generates executable DML statements, synthesizes approximate variants, executes them from the same database pre-state, and verifies their state-effect relations. We evaluate DMLScope on six widely used DBMSs and uncover 21 previously unknown logical bugs.

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